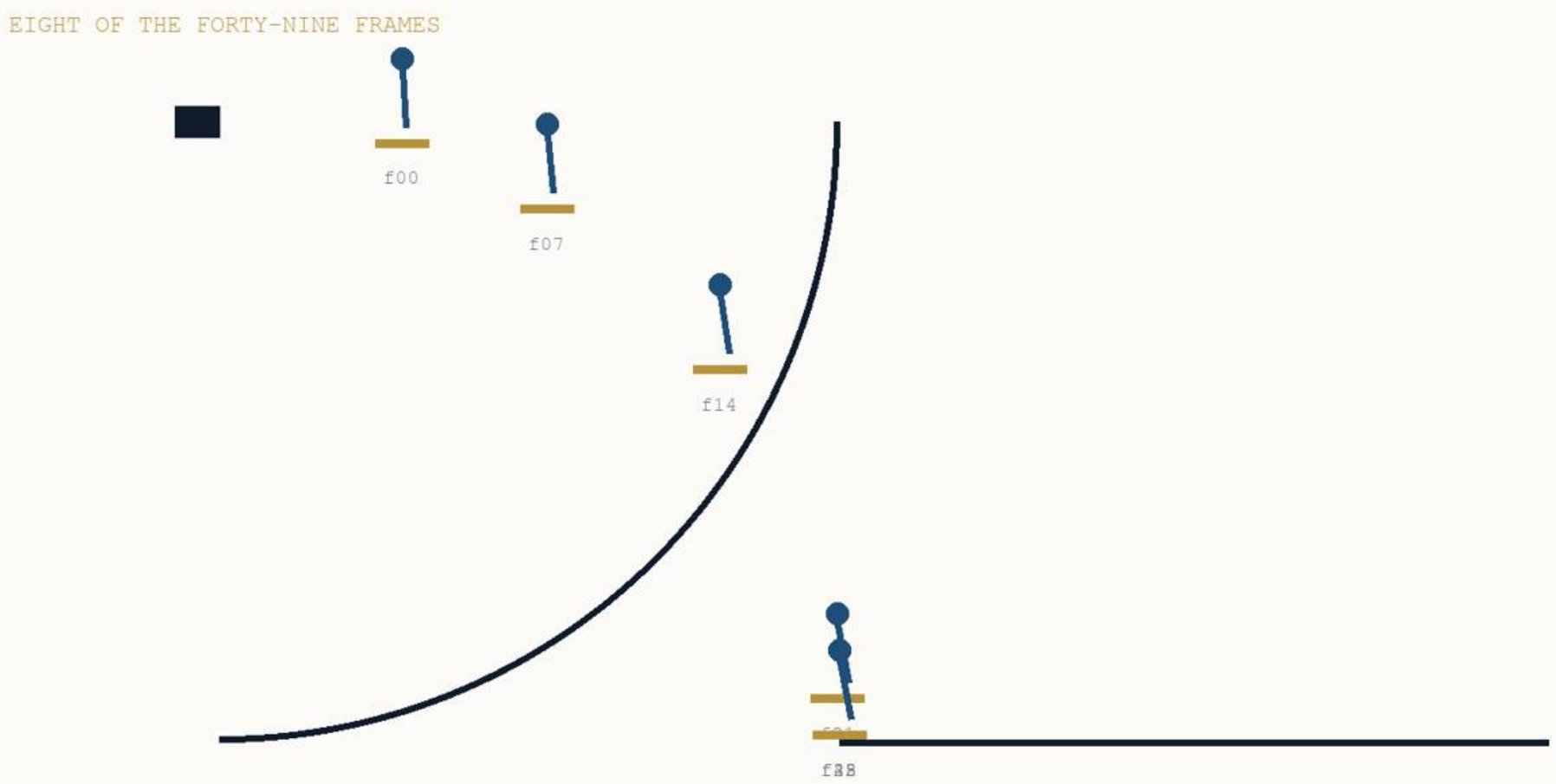
Drop-In at Washington Street

Rowan Hale · defended 9 April 2027

One drop-in on the 7-foot quarter at Washington Street, filmed at 30 frames a second, measured against the wall tiles, and explained with the calculus that governs it. The question the course asked on Day 1 was "changing how fast, right now?" This is the answer for 1.6 seconds of my own life.



Position read off the wall tiles (30 cm each); the phone was on a tripod at the top of the opposite wall.

FRAMES 0–48, EVERY THIRD FRAME SHOWN

The data

frame	t (s)	s (m)	v (m/s)	a (m/s²)
0	0.000	0.630	1.890	2.802
3	0.100	0.838	2.184	3.361
6	0.200	1.081	2.565	4.282
9	0.300	1.369	3.043	5.298
12	0.400	1.711	3.627	6.383
15	0.500	2.119	4.320	7.465
18	0.600	2.604	5.116	8.402
21	0.700	3.173	5.989	8.963
24	0.800	3.925	6.227	-0.936
27	0.900	4.541	6.134	-0.922
30	1.000	5.148	6.042	-0.909
33	1.100	5.747	5.952	-0.895
36	1.200	6.336	5.863	-0.882
39	1.300	6.916	5.776	-0.869
42	1.400	7.488	5.690	-0.856
45	1.500	8.051	5.605	-0.843
48	1.600	8.606	5.521	-0.834

Velocity is the central difference of position; acceleration the central difference of velocity. The first and last frames borrow their neighbours.

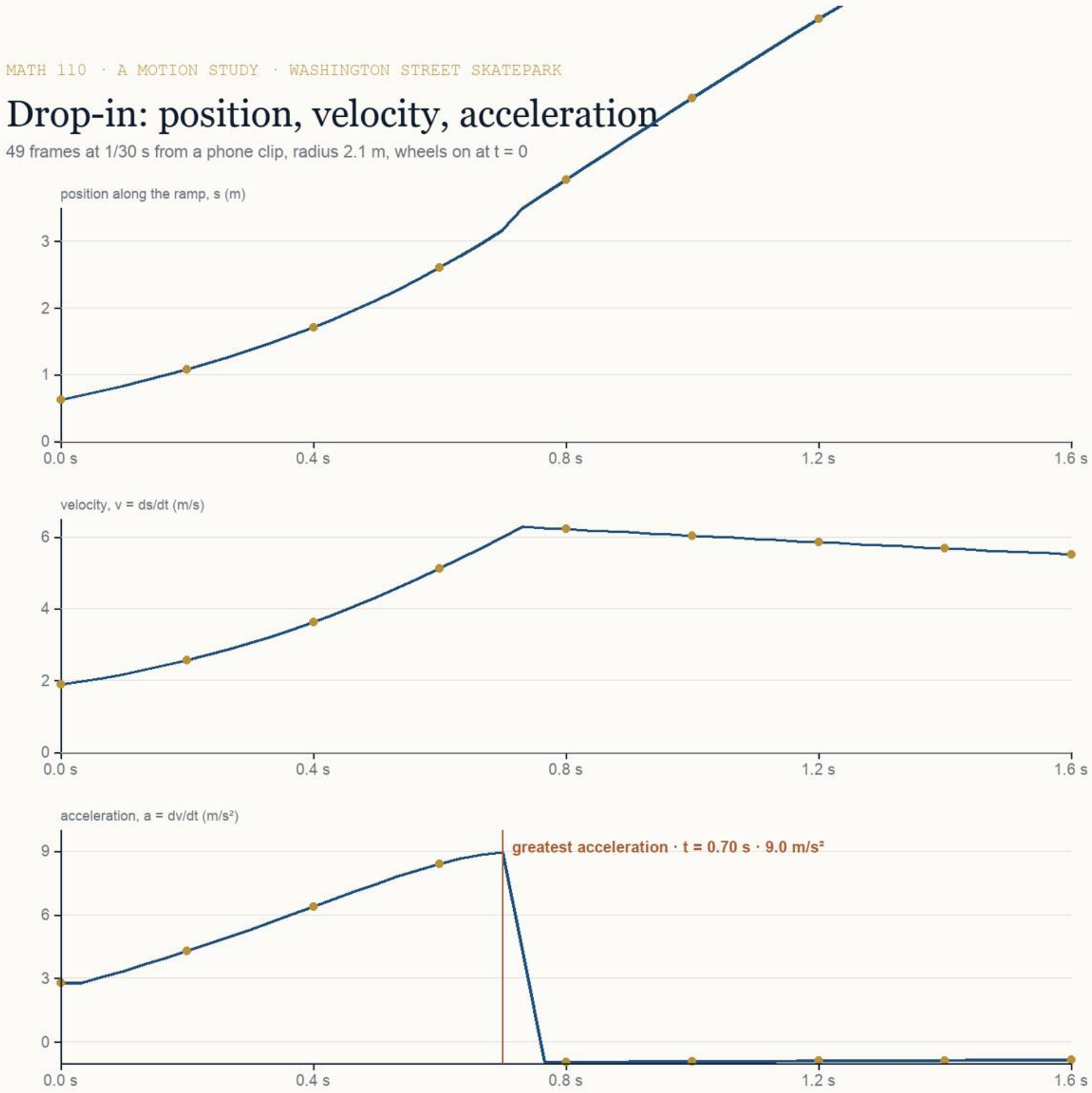
WHAT THE NUMBERS ARE DOING

The calculus

s(t) is the distance along the ramp. Its derivative, v(t) = ds/dt, is the slope of the position curve at each instant: the speed the wall tiles were going past. Its second derivative, a(t) = dv/dt, is how fast that speed was changing. The greatest acceleration came at t = 0.70 s, 9.0 m/s², about a third of the way down, where the ramp is steepest under the wheels and gravity's pull along the surface, g·sin θ, is largest.

By the flat the velocity reached 6.3 m/s and the acceleration fell toward zero: the slope was gone, so there was nothing left to change the speed except the wheels.

Energy check: ½v² at the bottom should equal g·h with h = 2.1 m, so v ≈ 6.4 m/s. The measured 6.3 m/s is lower by about what the wheels and the wind cost. The difference is the friction I could feel and now can also count.



The tangent line at each frame is the velocity. Where the velocity's own slope is steepest is where it felt like falling.